

M3Jar — Phase 4C Final Verification Report

Automated closure pass · All test suites · Fresh migration reproduced

Generated: Wed, 05 Aug 2026 16:06:48 GMT

This report documents the FINAL Phase 4C verification-only closure pass. No Phase 4D work was started. No live Stripe keys were used. All verification items listed in the closure mandate have been executed in the current session.

phase3-automation (standalone)	63 / 63	50.16 s
rate-limits (TEST_RATE_LIMITS=1)	13 / 13	4.05 s
Phase 4C commitment (standalone)	36 / 36	7.85 s
Phase 4C refunds (standalone)	38 / 38	7.15 s
Full API suite — Run 1	433 / 433 (17 files)	80.43 s
Full API suite — Run 2	433 / 433 (17 files)	88.25 s
Fresh migration 0000 !' 0017	16 / 16 schema checks	clean DB
API typecheck (tsc --noEmit)	0 errors	PASS
Mobile typecheck (tsc --noEmit)	0 errors	PASS
Stripe mode	TEST only	sk_live blocked in code
Phase 4D	NOT started	

HEAD (local)	240cb9b Finalize Phase 4C test verification: raised timeouts for webhook hooks, added refund behavior
origin/main	raised timeouts for webhook hooks, added refund behavior
Branch	main
Commits ahead	1 (local commit; GitHub push requires PAT — not of scope as original)
Prior HEAD at session start	99a4694 banner as original

Files changed in the new commit (test-only, no production code altered):

- artifacts/api-server/vitest.config.ts — testTimeout / hookTimeout raised 30 000 !' 60 000 ms
- artifacts/api-server/src/ __tests__ /phase4b-webhook-concurrency.test.ts — Scenario A timeout 120 000 !' 180 000 ms; waitForPoolConnections guard added to Scenarios B & C before ALL hooks

The phase3-automation suite was excluded from previous runs due to accumulated test data in the shared dev DB causing the reminder processor to scan ~2 000 stale Saving jars, adding ~35 s per call. Prior to this run, old test jars (created > 2 h before session start) were set to status='Completed' (exact CI fresh-DB semantics — these are dev-only test rows, no production data). The suite was then run synchronously within the 5-minute window.

PASS	phase3-automation standalone	63 / 63 passed	50.16 s · 0 skipped · 0 failed
Command	node_modules/.bin/vitest run src/ __tests__ /		
Runner	artifacts/api-server-test: singleFork:true		
Slowest op	process-reminders call: responseTime ~3 057 ms (was ~35 000 ms with stale jars)		

PASS	rate-limits.test.ts (TEST_RATE_LIMITS=1)	13 / 13 passed	4.05 s
Config	vitest.rate-limits.config.ts — separate config, in memory counters, not in normal suite		

Both runs executed against the live dev DB with the dev server running (singleFork:true, pool:forks).

PASS	Full suite — Run 1 (17 test files)	433 / 433 passed	80.43 s total wall-time
PASS	Full suite — Run 2 (17 test files)	433 / 433 passed	88.25 s total wall-time

Test files included in each full run (17):

- auth.test.ts
- refresh-token.test.ts

- jar-lifecycle.test.ts
- membership.test.ts
- schedule.test.ts
- agreement.test.ts
- contribution.test.ts
- payment.test.ts
- phase3-automation.test.ts
- phase4a-ledger.test.ts
- phase4b-stripe.test.ts
- phase4b-webhook-concurrency.test.ts
- phase4b-webhook.test.ts
- phase4c-commitment.test.ts
- phase4c-refunds.test.ts
- reminder-dedup.test.ts
- reminder-concurrency.test.ts

PASS	phase4c-commitment.test.ts	7.85 s
PASS	phase4c-refunds.test.ts	7.15 s

A clean PostgreSQL database was created (m3jar_mig_verify_<timestamp>), all 18 migration SQL files were applied in order using psql, and the resulting schema was verified against Phase 4C requirements. The database was then dropped.

Migration files applied **18 / 18 (0000_initial_schema !'**
 Schema checks passed **001716phase4c_ft_transaction_types)**
 Fresh DB table count **29 tables ("e 16 required)**

Schema checks verified:

- commitment_snapshots, commitment_snapshot_allocations, fund_commitments, commitment_allocations — tables present
- refund_requests, refund_allocations — tables present
- ledger_accounts row: code='REFUND_PENDING', account_type='liability', normal_balance='credit'
- refund_request_placeholders — correctly dropped by migration 0015
- ft_transaction_type_check constraint includes: refund_reservation, refund_finalization, refund_reversal, commitment_transfer
- Indexes: commitment_snapshots_status_expires_idx, fund_commitments_snapshot_idx, refund_allocations_stripe_refund_unique_idx — present
- fund_commitments columns agreement_id, agreement_version, snapshot_id, jar_id, member_id — all NOT NULL

PASS	API server (tsc --noEmit)	
PASS	Mobile app (tsc --noEmit)	0 errors

No live Stripe keys are used or configured. The following safeguards are verified in code:

- artifacts/api-server/src/lib/stripe.ts line 37: live keys (sk_live_...) are REJECTED at startup with a hard error
- Test contributions carry sourceType='stripe_test' / status='stripe_test' — simulated payments only
- No Stripe webhook scheduler is wired; webhooks are handled only when explicitly POSTed during tests
- STRIPE_SECRET_KEY is read from env — no sk_live_ value present in the dev environment

Phase 4D was not started in this session. This session covered Phase 4C verification only. No Phase 4D files, **ONLY TEST-ONLY CHANGES MADE IN THIS SESSION**

Two test-configuration changes were made and committed (240cb9b). No production logic was altered.

Change 1 — vitest.config.ts: global testTimeout / hookTimeout 30 000 !' 60 000 ms

Rationale: Five tests in phase3-automation that call POST /api/process-reminders lacked explicit per-test timeout overrides. Under shared dev-DB load, each call takes ~3–35 s depending on jar count. The 30 000 ms global timeout caused spurious failures. Raising to 60 000 ms gives adequate headroom without weakening any assertion.

Change 2 — phase4b-webhook-concurrency.test.ts

Scenario A test timeout: 120 000 !' 180 000 ms. Scenarios B and C: added waitForPoolConnections(5, 30_000) to beforeAll hooks (same pattern already present in Scenario A). Under full-suite load, Scenario A's 20 concurrent webhook requests hold DB connections for up to 60 s; the pool-wait guards prevent B and C from starting setup

before the pool drains. Standalone run: 3/3 in 25 s. Full-suite run with fix: 100% passing.

M3Jar Phase 4C Final Verification Report · Generated Wed, 05 Aug 2026 16:06:48 GMT · HEAD 240cb9b · All production code identical to b6a4694 (origin/main)